

Phase 7: Project Documentation Phase

1. Introduction

The Project Documentation Phase compiles all technical, functional, operational, and user-facing records for the **Smart Budget AI** platform. Comprehensive documentation ensures long-term maintainability, seamless onboarding for new developers, compliance with financial data standards, and effortless end-user operation.

2. Structure & Types of Documentation

Document Type	Target Audience	Primary Content / Purpose
System Architecture Document (SAD)	System Architects & Lead Developers	High-level component diagrams, tech stack specs, data flows, and microservice relationships.
API Reference Documentation	Backend & Third-Party Developers	Endpoint specifications, HTTP methods, request/response schemas, authentication headers, and status codes.
AI/ML Model Documentation	Data Scientists & ML Engineers	Model training pipelines, datasets used, feature engineering steps, classification metrics, and accuracy benchmarks.
Database Schema & Data Dictionary	Database Administrators (DBAs)	ER diagrams, table definitions, indexing strategies, field data types, and primary/foreign key constraints.
User Manual & Help Guide	End-Users & Non-Technical Stakeholders	Step-by-step guides for setup, expense logging, automated budgeting, notification preferences, and troubleshooting.
Deployment & DevOps Guide	DevOps & System Administrators	CI/CD pipeline steps, environment variables, Docker setup, cloud hosting configuration (AWS/GCP), and monitoring procedures.

3. Key Technical Specifications

3.1 Tech Stack Summary

- **Frontend:** React.js / React Native (Mobile & Web UI)
- **Backend:** Node.js (Express.js) & Python (FastAPI for Machine Learning services)
- **Database:** PostgreSQL (Transactional financial data) & Redis (Caching & Session management)

- **AI Engine:** Scikit-Learn / PyTorch (NLP for expense categorization & time-series forecasting for budget predictions)
- **Security & Auth:** OAuth 2.0, JWT (JSON Web Tokens), AES-256 Encryption (At Rest), TLS 1.3 (In Transit)

3.2 Sample API Endpoint Specification

- **Endpoint:** POST /api/v1/transactions/categorize
- **Description:** Accepts a transaction description and returns an AI-predicted expense category with confidence scores.
- **Request Payload (JSON):**

```
{
  "user_id": "usr_98765",
  "amount": 42.50,
  "currency": "USD",
  "raw_description": "Walmart Supercenter"
}
```

- **Response Payload (JSON):**

```
{
  "status": "success",
  "predicted_category": "Groceries",
  "confidence_score": 0.96,
  "budget_status": {
    "allocated": 500.00,
    "remaining": 185.20,
    "exceeded": false
  }
}
```

4. Security, Governance & Compliance Standards

- **Data Privacy:** Full compliance with data protection policies (e.g., GDPR / local data protection guidelines) ensuring financial user logs are anonymized.
- **Encryption Standards:** Financial credentials and banking tokens are encrypted using AES-256 bits before database entry.
- **Audit Logs:** Immutable audit trails for critical operations such as password changes, account deletions, and budget threshold updates.

5. Maintenance & Version Control Guidelines

- **Repository Strategy:** Managed via Git with protected main and develop branches; code additions require passing peer review and automated unit tests.
- **Versioning Protocol:** Semantic Versioning (v1.0.0 - Major.Minor.Patch) applied across releases.
- **Documentation Maintenance:** API specs automatically updated via Swagger/OpenAPI tools upon code deployment.

6. Deliverable Summary

The complete documentation suite for **Smart Budget AI** is compiled and archived in the project repository under /docs. The system is fully documented and prepared to proceed to **Phase 8: Project Demonstration Phase**.